

2.7.23

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Question

For what value of k ($k > 0$) is the area of the triangle with vertices $(-2, 6)$, $(2k, 4)$ and $(2k + 1, 10)$ equal to 52 sq. units?

Solution

Vertices of triangle are

$$\mathbf{A} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} \quad (1)$$

$$\mathbf{B} = \begin{pmatrix} 2k \\ 4 \end{pmatrix} \quad (2)$$

$$\mathbf{C} = \begin{pmatrix} 2k + 1 \\ 10 \end{pmatrix} \quad (3)$$

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} -2 - 2k \\ 6 - 4 \end{pmatrix} = \begin{pmatrix} -2 - 2k \\ 2 \end{pmatrix} \quad (4)$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} -2 - (2k + 1) \\ 6 - 10 \end{pmatrix} = \begin{pmatrix} -2k - 3 \\ -4 \end{pmatrix} \quad (5)$$

$$(\triangle ABC) = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| \quad (6)$$

$$(\triangle ABC) = \frac{1}{2} \left\| \begin{pmatrix} -2 - 2k \\ 2 \end{pmatrix} \times \begin{pmatrix} -2k - 3 \\ -4 \end{pmatrix} \right\| = \frac{1}{2} |12k + 14| \quad (7)$$

Given area of triangle = 52 sq.units,

$$\frac{1}{2} |12k + 14| = 52 \quad (8)$$

$$|12k + 14| = 104. \quad (9)$$

Solution

Now consider the two cases from the absolute value,

Case 1:

$$12k + 14 = 104 \quad (10)$$

$$12k = 90 \quad (11)$$

$$k = \frac{90}{12} = \frac{15}{2}. \quad (12)$$

Case 2:

$$12k + 14 = -104 \quad (13)$$

$$12k = -118 \quad (14)$$

$$k = -\frac{118}{12} = -\frac{59}{6}. \quad (15)$$

Because the problem requires $k > 0$, we discard the negative solution.

$$k = \frac{15}{2}$$

Code (C)

Part 1

```
#include <stdio.h>

#ifdef _WIN32
#define EXPORT __declspec(dllexport)
#else
#define EXPORT
#endif

// Function to compute the positive value of k
EXPORT double solve_k_positive() {
    double k1 = (104.0 - 14.0) / 12.0; // 7.5
    double k2 = (-104.0 - 14.0) / 12.0; // -9.8333

    return (k1 > 0) ? k1 : k2;
}
```

Part 2

```
int main() {  
    double k = solve_k_positive();  
    printf("Positive value of k: %.2lf\n", k);  
    return 0;  
}
```

Python Code:Import and Define points

Part 1

```
import ctypes
import os
import matplotlib.pyplot as plt

# Load compiled shared library from code4.c
if os.name == "nt": # Windows
    lib = ctypes.CDLL("./code4.dll")
else: # Linux/macOS
    lib = ctypes.CDLL("./code4.so")

# Define C function signature
lib.solve_k.argtypes = [ctypes.POINTER(ctypes.c_double)]
lib.solve_k.restype = None
```


Part 2

```
# Prepare array for results
results = (ctypes.c_double * 2)()
lib.solve_k(results)

# Extract values of k
k1, k2 = results[0], results[1]
print(f"Possible values of k: {k1:.2f}, {k2:.2f}")

# Pick valid solution (k > 0)
k = k1 if k1 > 0 else k2
print(f"Chosen k = {k:.2f}")
```

Part 4

```
# Define triangle vertices
A = (-2, 6)
B = (2 * k, 4)
C = (2 * k + 1, 10)

# Plot triangle
plt.figure(figsize=(7,7))
plt.plot([A[0], B[0], C[0], A[0]],
         [A[1], B[1], C[1], A[1]],
         'b-', linewidth=2)

# Plot vertices as red points
plt.scatter([A[0], B[0], C[0]], [A[1], B[1], C[1]],
           color='red', s=80, zorder=5)
```

Python Code: Plotting

Part 5

```
# Annotate vertices with coordinates
```

```
plt.text(A[0]+0.2, A[1]+0.2, f"A{A}", fontsize=12)
```

```
plt.text(B[0]+0.2, B[1]-0.4, f"B{B}", fontsize=12)
```

```
plt.text(C[0]+0.2, C[1]+0.2, f"C{C}", fontsize=12)
```

```
plt.title(f"Triangle with k = {k:.2f}")
```

```
plt.grid(True)
```

```
plt.axis("equal")
```

```
# Save as fig4.png
```

```
filename = "fig4.png"
```

```
plt.savefig(filename, dpi=200)
```

```
print(f"Figure saved as {filename}")
```

```
# Show the figure
```

```
plt.show()
```

Graphical Representation

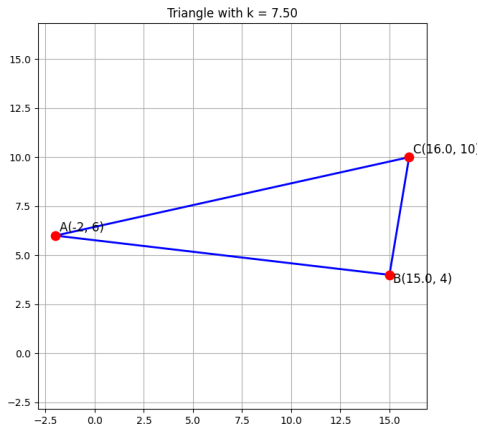


Figure: Area of triangle